Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Customer and purchase details

37

Missing Values

Review ratings imputed

Key Features

- Demographics: Age, Gender, Location
- Purchase details: Category, Amount, Season
- Behavior: Discounts, Promo codes, Frequency

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Data Preparation Process

Data Loading

Imported using pandas

Initial Exploration

Structure and statistics check

Missing Data

Median imputation by category

Feature Engineering

Age groups, frequency calculations

Database Load

PostgreSQL integration

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SQL Analysis: Key Findings

Revenue by Gender

Male: $157,890

Female: $75,191

Discount Users

839 customers used discounts above average purchase

Top Products

Gloves (3.86), Sandals (3.84), Boots (3.82)

Shipping Comparison

- Standard: $58.46 avg
- Express: $60.48 avg

Discount-Dependent Products

- Hat: 50%
- Sneakers: 49.66%
- Coat: 49.07%

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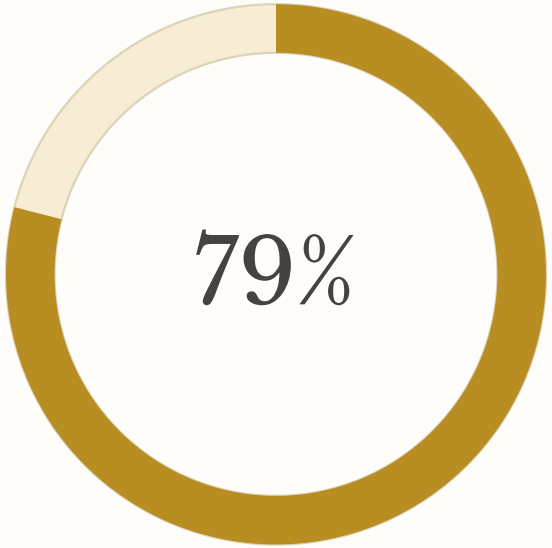
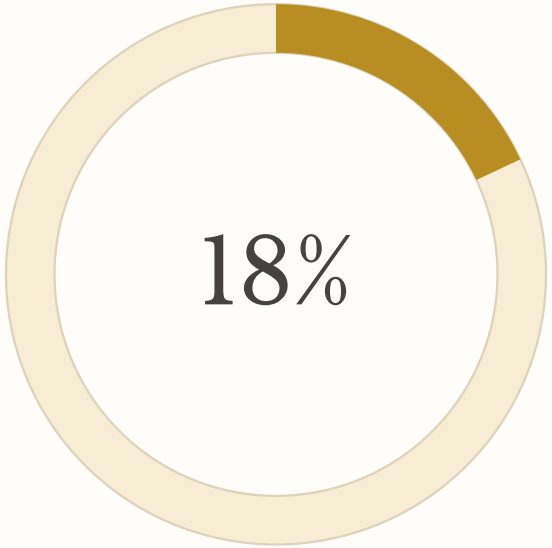
Customer Segmentation



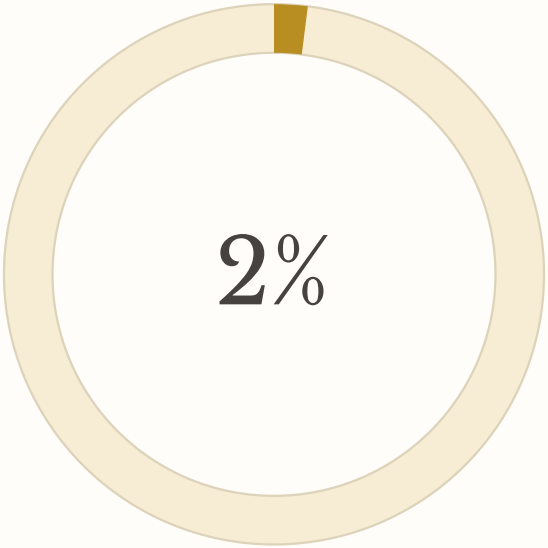
Loyal Customers

3,116 customers with repeat purchases



Returning

701 customers making second purchases



New

83 first-time buyers

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Product Performance by Category

1	**Clothing** Blouse, Pants, Shirt 1,800 sales | $100K revenue
2	**Accessories** Jewelry, Sunglasses, Belt 1,200 sales | $70K revenue
3	**Footwear** Sandals, Shoes, Sneakers 600 sales | $30K revenue
4	**Outerwear** Jacket, Coat 300 sales | $20K revenue

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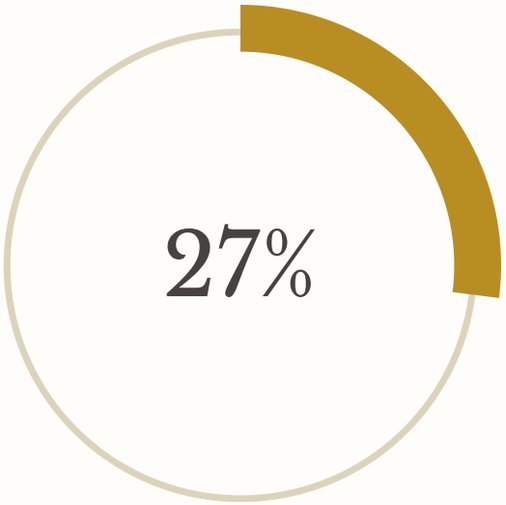
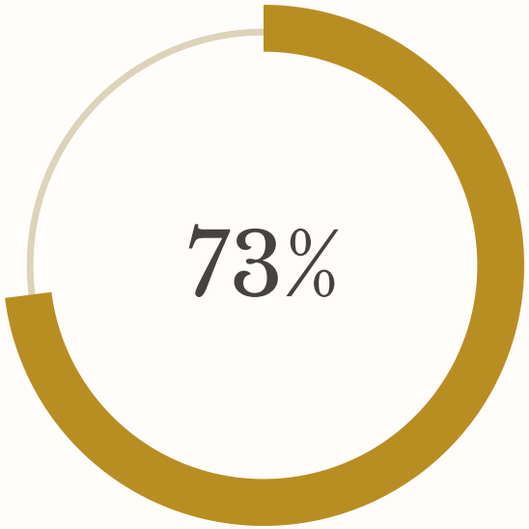
Subscription Insights



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

Revenue Comparison

Subscribers: $62,645 total

Non-subscribers: $170,436 total

Key Finding: Repeat buyers (over 5 purchases) show higher subscription rates

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Revenue by Age Group

27%

Young Adult

$62,143 revenue | 1,000 sales

24%

Middle-aged

$59,197 revenue | 900 sales

22%

Adult

$55,978 revenue | 800 sales

27%

Senior

$55,763 revenue | 700 sales

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Key Business Insights

Gender Gap

Male customers generate 2.1x more revenue than female customers

Discount Strategy

High-discount products (Hat 50%, Sneakers 49.66%) drive sales but impact margins

Product Quality

Gloves, Sandals, Boots have highest ratings (3.82-3.86)

Shipping Preference

Express shipping customers spend 3.5% more ($60.48 vs $58.46)

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Strategic Recommendations

01

Boost Subscriptions

Promote exclusive benefits and rewards for subscribers

02

Enhance Loyalty Programs

Reward repeat buyers to accelerate movement into loyal segment

03

Optimize Discount Policy

Balance sales boosts with margin control across product categories

04

Strategic Product Positioning

Highlight top-rated and best-selling products in marketing campaigns

05

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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